

2026-04-29 Project Background with Architect — AWS EBS Snapshots for SQL Server Backups

2026-04-29 — Project Background with Architect — AWS EBS Snapshots for SQL Server Backups

⚠ AI-generated document

This document was generated entirely by AI from a source VTT transcript file. It has not been manually reviewed. Any quoted figures, costs, or metrics are estimated from participants' verbal recall and are subject to confirmation.

Attendees	Dave Nelson, Ross Bell, Becky Boardman, Daniel Kaminski, Neil Romatowski, Olufunwa, Angela
Duration	~54 minutes
Source	[PROCESSED] [Project Background with Architect] AWS EBS Snapshots for SQL Server backups.vtt

TL;DR

The DBA team briefed infrastructure architect Dave Nelson on the N2W backup replacement project — a POC to replace NetBackup with EBS snapshot-based backups (via N2W) and transaction log storage on FSx for ONTAP, aiming to reduce RTO from 3 working days to under 4 hours, RPO from 9 hours to 15 minutes, and potentially halve the ~\$1.5M backup costs.

Decisions, action items, and ownership below are captured from the AI extraction only. Actual status and assignments are tracked via the ADO Kanban board. Ownership is AI-inferred from speech patterns unless a participant explicitly committed — confirm assignments on the board.

Decisions

- **D1 (Accepted):** The DBA team will lead solution architecture for the N2W backup replacement; the architecture team (Dave Nelson) will provide review and oversight on request. Owner: Becky Boardman.
- **D2 (Accepted):** The initial scope is a proof of concept, not a production-ready solution. The POC will use ClickOps; infrastructure as code is deferred to a later phase. Owner: Ross Bell.
- **D3 (Accepted):** Dave Nelson will close the duplicate AHA demand (assigned by Nicola Richardson) and reference the existing one in the Plateng space. Owner: Dave Nelson.

- **D4 (Accepted):** Transaction log backups will use 15-minute intervals with 14-day retention, stored on FSx for ONTAP with automatic hot/cold tiering and cross-AZ DR replication. Owner: Ross Bell.
- **D5 (Accepted):** The POC RPO target is up to 15 minutes of potential data loss, reduced from the current 9-hour window. Owner: Ross Bell.
- **D6 (Proposed):** Cost attribution via resource tagging will be deferred to phase 2 or 3 — not part of the POC. Blocker: POC must complete first. Owner: Ross Bell.
- **D7 (Accepted):** The project restart will begin with 1–2 weeks of project management setup (documentation location, meeting cadence, stand-ups) before technical implementation. Owner: Ross Bell.

Action items

Owner	Action	Due	Status
Dave Nelson	Close duplicate AHA demand and reference existing Plateng demand	—	Open
Becky Boardman	Send AHA demand link to Dave Nelson via chat	—	Open
UNASSIGNED	Send N2W recovery requirements document to Dave Nelson	—	Tentative
UNASSIGNED	Conduct requirements gathering session with Neil Romatowski's team when project restarts	—	Tentative

Open questions

- What are the contractual RTO and RPO obligations with the customer? Ross Bell raised this — asked a year ago and nobody could answer. Dave Nelson agreed this is critical to establish. Owner: UNASSIGNED.
- Has N2W committed to developing the SQL Server application-aware restore feature (restoring state support)? Olufunwa, Angela noted there is no firm commitment from either side yet. Owner: UNASSIGNED.
- When will the Group DBA licence remediation P1 project complete, freeing DBA team resources for N2W? Owner: Becky Boardman.

Discussion

Project background and the XY problem

Ross Bell framed the project using the XY problem analogy: an earlier solution (now discarded as irrelevant) was abandoned after the team recognised it was solving the wrong problem. The actual problems are a 3 working-day RTO per node per patch domain and up to 9 hours of potential data loss. The project is a proof of concept with phased delivery — the hope is the POC becomes an MVP that can be deployed. Becky Boardman noted the DBA team has had limited traction due to competing demands on other projects but expects capacity to free up soon.

Architecture team engagement and duplicate AHA demand

Dave Nelson arrived with minimal context — an AHA demand (assigned by Nicola Richardson) with sparse details reading "N2W POC backup and cost reduction." Becky Boardman identified this as a duplicate of an existing demand tracked in the Plateng space. Dave Nelson agreed to close the duplicate. Gary had separately told Dave Nelson that he was "doing the designing costing for the transaction logs using NetApp FSX with the DBA team," which both Becky Boardman and Ross Bell noted was inaccurate — that work was already done.

Current backup costs and savings potential

Ross Bell noted that NetBackup costs approximately \$1.5 million *[Unverified]* to run and suggested the N2W solution could potentially more than halve that cost *[Unverified]*. Neil Romatowski raised the need to quantify savings to present to the business. Ross Bell confirmed the team has already produced cost estimates with technical backing — the FSx for ONTAP solution is estimated at \$13,000 per month including storage, replication, and DR standby costs.

N2W vendor engagement and feature gap

Ross Bell explained that N2W currently cannot restore SQL Server databases to a restoring state — it fully recovers the database, which prevents applying transaction log backups on top. The team engaged N2W with a document containing four essential recovery requirements. N2W indicated willingness to

implement the feature and suggested it would not be difficult. Becky Boardman noted the team is offering a lucrative business opportunity, and if N2W develops this feature, other customers would benefit. Olufunwa, Angela confirmed N2W understands the requirements but noted no firm commitment exists from either side.

Cost attribution and tagging for TCO

Neil Romatowski raised the need to segregate costs by business service (e.g., GP Live, CCMH) for total cost of ownership reporting, suggesting resource-level tagging baked into code from the design stage. Ross Bell noted that N2W has a cost appropriation module using tagging for snapshot backups. For the FSx for ONTAP transaction log storage, the team can encode business domain into file paths and derive cost splits as percentages. Ross Bell proposed deferring this to a later phase — the POC will use ClickOps, not infrastructure as code. Neil Romatowski agreed this approach is reasonable.

Key technical discussions

EBS snapshots and volume-level recovery

N2W orchestrates AWS EBS snapshot backups with intelligent tiering policies, moving snapshots to Glacier storage automatically. EBS volume restoration is near-instant (seconds), though end-to-end recovery involves additional steps. Ross Bell cited prior experience at a previous company using Cohesity (on-prem volume snapshots) where full servers with large databases were recovered in 40–50 minutes. The POC target is under 3–4 hours RTO, down from the current 3 working days.

FSx for ONTAP architecture for transaction log backups

SQL Server 2019 cannot backup direct to S3 (only SQL Server 2022 supports this). The team selected FSx for ONTAP as a central network share for all transaction log backups. It provides automatic hot/cold tiering: ~2 days on SSD hot tier, then automatic migration to the capacity pool (cold tier). A DR standby in another AZ replicates automatically and fails over if needed. The team member Rob has prior experience with the on-prem ONTAP solution. Unlike S3 storage classes, FSx for ONTAP tiering does not incur the retrieval penalties that S3 Standard-Infrequent Access has for early access.

RDS and managed instance alternatives

The architecture team has had extensive conversations about alternatives to running SQL Server on Windows VMs in the cloud — including RDS managed instances, Linux-based SQL, and Babelfish (Aurora compatibility layer). Babelfish was evaluated but rejected by John McCluskey due to technical debt in stored procedures. Darren Cheevles is now exploring stored procedure optimisation to enable CPU right-sizing. Dave Nelson noted this is a top architecture focus alongside GPX.

Breakthrough moments

- **Ross Bell** reframed the entire project through the XY problem analogy, demonstrating that an earlier discarded solution was solving the wrong problem. The core issues are the 3 working-day RTO and 9-hour RPO — not a specific technology choice. This reframing led to the current N2W-based architecture with EBS snapshots for recovery time and FSx for ONTAP for recovery point. *[Inferred]*

Outcomes

Dave Nelson arrived with near-zero context about the N2W project. He now understands the core disaster recovery problems (3-day RTO, 9-hour RPO), the proposed two-part solution (N2W EBS snapshots for full backups plus FSx for ONTAP for transaction log storage), the estimated costs (\$13,000/month for FSx), the vendor engagement status with N2W, and the phased delivery approach. The duplicate AHA demand was identified and resolution agreed. Neil Romatowski's cost attribution requirements were acknowledged and slotted for a future phase.

Forward look

- The DBA team plans to restart the N2W project once the Group DBA licence remediation P1 project completes, targeting the next few weeks.
- The first 1–2 weeks of restart will focus on project management setup — documentation storage, meeting cadence, and stand-ups — before technical implementation begins.
- A mobbing session using Facilitator will be spun up to track the project, capture key decisions, and enable stakeholders to review progress asynchronously.
- Dave Nelson offered to have informal early-morning conversations about SQL cost optimisation with Ross Bell when both are online.

Architecture decisions

ADR-draft: Use N2W with EBS snapshots and FSx for ONTAP for SQL Server backup and recovery

- **Status:** Proposed
- **Context:** Current disaster recovery takes 3 working days per node (RTO) with up to 9 hours potential data loss (RPO). NetBackup costs approximately \$1.5M *[Unverified]*. Building an in-house backup and recovery solution is not viable. N2W orchestrates EBS snapshots with tiering policies but currently lacks application-aware SQL Server restore capability (restoring state).
- **Decision:** We will use N2W for EBS snapshot-based full backups with tiering, and FSx for ONTAP for transaction log backup storage (15-minute intervals, 14-day retention, SSD hot tier for ~2 days with automatic cold tier migration, cross-AZ DR replication with automatic failover).
- **Consequences:**
 - (+) RTO reduced from 3 working days to potentially under 1 hour

- (+) RPO reduced from 9 hours to 15 minutes
 - (+) Potential >50% reduction in backup costs *[Unverified]*
 - (+) Automatic cross-AZ DR failover for transaction log storage
 - (–) Dependency on N2W vendor to develop application-aware restore feature (restoring state support) before the full solution is viable
 - (–) SQL Server 2019 cannot backup direct to S3, requiring FSx for ONTAP as intermediate storage — a constraint that may be eliminated when the estate fully upgrades to SQL Server 2022
-